

Data Structures Interview Questions :

1. What is Data Structure ?

Data Structure is a particular way of storing & organizing data in computer's memory so that the data can be efficiently accessed.

3. What are the various types of DS ?

There are two types of data structures :-

- Primitive DS
- Non - Primitive DS

Primitive data structures are the data structures which can store only single value.

Ex → int, float, bool, pointer, etc.

Non-primitive data structures are those data structures which are derived from primitive data structures. And they are ~~then~~ further categorized into two types —

- Linear DS.
- Non-Linear DS.

4. What is Linear & Non-Linear Data structure?

Those data structures in which one element is only connected to one other element are called Linear DS.

Ex → Array, Queue, stack, structure & Linked List.

While, Non-Linear data structures are those data structures in which one element is connected to multiple elements.
Ex → Tree & Graph.

5. what is Array?
6. what is Structure & How is it diff. from Array?
7. what are the various Advantages and disadvantages of using an Array?
The best advantage of using an array is ~~that~~ accessing speed. In array, we can access any element in constant amount of time with time complexity of $O(1)$. [big oh of 1]

While, if I talk about the disadvantage.
So, it has static memory allocation, that's why we cannot increase or decrease its size at run-time.

Secondly, Insertion & deletion are costly operations in Array.

for ex,

To insert any element in the array, we first have to shift all the elements after it one position right. That cost $O(n)$ time complexity.

And,

To delete any element, we have to shift all the element ^{after} it one position left. This also cost $O(n)$ time complexity.

8. Diff. btw Array & Linked List ?

Array has static memory allocation. That means we cannot increase or decrease its size at runtime.

Secondly, Insertion & deletion are the costly operations in array i.e. they cost $O(n)$ time complexity.

While, In Linked List, we have Dynamic memory allocation that's why the size of Linked List increase / decrease at runtime.

Secondly, Insertion & deletion are very easy operations in Linked List. They just ~~are~~ have time complexity of $O(1)$.

9. What is stack & on which principle it is based ?

Stack is the Linear List with the restriction that elements will be inserted & deleted only from one end of the stack i.e. the top of the stack.

And, stack is based on the principle of LIFO i.e., Last In First Out.

10. what are the operations associated with stack?

There are three operations associated with stack that are:

push, pop, ~~pre~~ peek.

Now, push means to insert the element.

pop means to delete the element.

peek means to just see the topmost element.

11. what is the condition for stack overflow & stack underflow.

The condition for stack overflow is that top equals to maxIndex.

while,

The condition for stack underflow is top equals to -1.

12. Can you give any real-life example of stack?

Sir, ~~the~~ A marker is the best example of stack.

when we pour ink in the whiteboard marker, it starts going down but when we close the cap of the marker & start writing then the drop that inserted

at last will be utilized first. That's the wonderful example of stack.

13. what are the various applications of stack.

There can be multiple applications of stack. First of all, stack can be used to check whether the string is palindrome or not.

Secondly, it can be used for function calls.

Thirdly, it can be used to evaluate postfix expressions.

Fourth, it can be used for balancing the parentheses ~~brackets~~; that means to check whether the brackets / parentheses are balanced or not in a program.

14. what are expressions & How many expressions are there?

The expressions are nothing but the mathematical statements. They are of three types — prefix, infix & postfix.

In the prefix expressions, the operators come at the first position & operands come on second & third.

In the Infix expressions, the operator come in-between & ~~left~~ ~~&~~ ~~right~~ ~~come~~ ~~at~~ the ~~opera~~ operands come at left & right.

In the postfix expressions, the operator come at last & the operands come at first & second position.

15. what is Queue & On which principle it is based?

Queue is a linear list with the restriction that elements will be inserted from one end that is called as the rear end & deleted from another end called as the front end of the Queue.

And, it is based on the principle of FIFO, First In First Out.

16. what are the various operations associated with Queue?

There are two operations associated with the Queue which are:

Enqueue and Dequeue.

- Enqueue means to insert an element.
- Dequeue means to delete an element.

17. Can you give any real-life example of queue?

yes sir, A station

A ticket counter is a wonderful example of queue where people are standing in the line.

Another example can be car wash line.

18. Technical example of queue?

Sir, A printer can be a technical example of queue b'cuZ when many people given print request simultaneously then their requests are generated in a queue.

Another example can be CPU scheduling.

19. Diff. btw Stack & Queue.
Defination.

20. what is the condition for Queue Overflow & underflow.

The condition for Queue overflow is rear equals to maxSize.

while, The condition for Queue underflow is front equals to rear.

21. what is linked list & How many types of linked list are there?

Linked list is the collection of nodes in which each node contains two fields. The first one is the data field which contains the data & the second one is the link field which contains the address of the next node.

There are 3 types of Linked list :

- Singly
- Doubly
- Circular

22. Can you explain each type of linked list?

Yes sir, there are 3 types of Linked list. First of all, Singly linked list. So, it is the collection of nodes in which

Second one is Doubly linked list. So, it is also the collection of nodes but each node contains 3 fields. The middle field is a data field and the left & right side of the fields are called link fields. Right link contains the address of right node & left link contains the address of left node.

And, In circular linked list, the last node contains the address of first node.

23. what is linear search & what is its time complexity?

Linear search is a searching technique in which we search for the data linearly. we start from the 0th index & we go to the last index. And, the time complexity of linear search is $O(n)$.

24. what is Binary Search?

Binary search is a searching technique which works on sorted data and its time complexity is $O(\log n)$.

Now, Binary search works in a manner that, There are 3 variables, start, mid & end. So, start will always set to zero. mid will set to $\frac{\text{start} + \text{end}}{2}$ & end will set to

last index. Now, we start searching from the mid index. If the element is found on the mid index then we say that the element is found & we will break from the loop.

P.T.O

But, the element to be found is greater than the mid element then we set $start = mid + 1$ and $mid = \frac{start + end}{2}$.

And, if the element to be found is lesser than the mid element then, we set $end = mid - 1$ & $mid = \frac{start + end}{2}$.

We keep on doing this until $start$ is lesser than or equals to end . If in any case $start$ is greater than end then we will break from the loop.

25. what is the necessary conditions to perform Binary Search?

The necessary condition to perform Binary Search is that the array on which we are performing binary search should be sorted.

26. what is Sorting?

Sorting is a technique of sort / arranging the data in a specific ~~manner~~ ^{order}. It can be ~~a~~ ~~increas~~ ascending or descending order.

There are various sorting techniques available like : Bubble sort, radix sort, heap sort, bucket sort, etc.

27. what is a Tree or Binary Tree ?

A tree is a collection of Nodes & Edges .

And, a tree is said to be a binary tree if any of its node atmost two child nodes i.e any node of the binary tree cannot have more than two children .

28. what is Perfect , strict & complete B.T ?

A tree is said to be a perfect Binary tree if all of its levels are completely fulfilled .

Second is Strict Binary Tree . So , A tree is said to be strict binary tree , if all of its nodes have either zero or two children i.e any node cannot be there which have one or more than two children .

Third one is complete binary tree . So , A tree is said to be complete binary tree if it is a perfect binary tree & even if its last level is not fulfilled, all the extra nodes must be as left as possible .

29. what is Binary Search Tree ?

A tree is said to be Binary Search Tree, if all the nodes which have greater value than the root node goes to the right side of the root node & all the nodes which have lesser value than the root node goes to the left side of the root node.

30. what is tree traversing & How many types of tree traversal are there ?

Tree traversal means visiting each node of the tree exactly once without making any change.

There are two types of traversals :

- The first one is BFS which stands for Breadth First search.
- And the second one is DFS which stands for Depth First search.

In BFS, we traverse the tree level by level. First of all, we traverse zero level, then first, then second & so on.

And, In DFS, they are of 3 types :
Pre order, Inorder & Postorder.

Now, In the Pre-order traversal, we first visit the root, then left, then

right.

In the In-order traversal, we first visit left, then root, then right.

And, In the post-order traversal, we first visit ~~right~~ left, then right, then root.

31. what is AVL Tree & what is Balancing factor?

AVL tree is a height balancing Binary tree & In the AVL tree, the balancing factor of each node can be maximum to maximum 0, 1, -1.

Balancing factor = Height of left sub tree - Height of right sub tree.

32. Diff. btw Tree & Graph.

- A tree is a collection of nodes & edges while A graph is a collection of vertices & edges.
- A tree cannot have cycles but A graph can have cycles.
- Every tree is a graph but every graph is not a tree.

33. Can you give any real-life example of tree & where we use it in programming?

So, A ~~life~~ Real-life example of a tree data structure can be our family. In the first place, our grandfathers & grandmothers are there, then some children are there & some children of children are there. So, family is the best example of tree D.S.

And, if I talk about any technical example, so, operating system's file directory are the example of Tree D.S.

34. What is Graph & where we use Graph? Graph is the collection of vertices and edges & there are various applications of the Graph like Google maps, World wide web and finding mutual friends on social media.

35. What is strongly connected Graph? A Graph is said to be a strongly connected graph, if all its vertices are directly connected to other vertices of the graph.